

Lipika Sau

📍 Kolkata — ✉ misslipika.sau@gmail.com — ☎ +91 9064073327 — LinkedIn — GitHub

Professional Summary

Computer Science undergraduate (B.Tech 2022–2026) with hands-on experience in backend development, application troubleshooting, REST APIs, SQL, relational databases, authentication, and automated testing. Experienced in debugging frontend, backend, API, and database issues, developing reliable application components, validating data, and integrating multiple technologies. Strong analytical and problem-solving skills with practical experience using Python, JavaScript, SQL, Node.js, Express, FastAPI, PostgreSQL, and Git. Detail-oriented, adaptable, and motivated to build a career as an Associate System Engineer.

Education

University of Engineering and Management

2022 – 2026

B.Tech in Computer Science and Technology

CGPA: 7.64

Projects

FooCal – AI-Powered Food Calorie Tracker

[Github Link](#)

- Developed a full-stack application using Node.js, Express, PostgreSQL, and REST API integration, gaining hands-on experience in backend services, application development, and relational databases.
- Implemented JWT-based authentication with bcrypt password hashing for secure user login and session handling.
- Built role-based access control with an admin account seeding/promotion flow for managing user privileges.
- Integrated Google's Generative AI (Gemini) API to process user-uploaded food images and generate calorie estimates.
- Created meal history and insights functionality to retrieve, organize, and display application data through a structured interface.
- Debugged and integrated frontend, backend, API, and database components to identify and resolve application issues.

Customer support + SLA desk

[Github Link](#)

- Architected a full-stack SLA-driven support ticketing system using FastAPI (async), PostgreSQL, and SQLAlchemy, gaining hands-on experience in backend systems, relational databases, and application workflows.
- Developed automated background processing using Celery and Redis for scheduled tasks and SLA breach escalation.
- Designed and implemented an SLA engine to calculate business-hour-aware response and resolution deadlines, dynamically adjust to priority changes, and detect and escalate breaches.
- Built JWT-based role-access authentication for customer, agent, manager, and admin roles, securing REST APIs for ticket lifecycle management, administration, and reporting.
- Developed a responsive React dashboard with TanStack Query featuring real-time SLA countdown indicators and manager analytics for SLA compliance and agent performance.
- Ensured application reliability through automated unit and integration testing, including edge-case testing for weekends and after-hours using Freezegun.
- Debugged and integrated application components across the API, database, backend, and frontend layers to identify and resolve technical issues.

Certifications

Web Development Fundamentals

[View Certificate](#)

Database Structures and Management with MySQL

[View Certificate](#)

Python for Data Science, AI & Development

[View Certificate](#)

Technologies

Programming & Database : Python, JavaScript, SQL, MySQL, PostgreSQL, Relational Database Management, Database Structures, Data Querying, Data Validation, Object-Oriented Programming

Web Technologies : HTML5, CSS3, Tailwind CSS, Vanilla JavaScript, Responsive Design, Single-Page Applications, Cross-Browser Compatibility

Back-end & Integration : Node.js, Express, REST APIs, API Integration, JSON, Backend Development, Authentication, Role-Based Access Control

Tools & Workflow : Git, GitHub, Redis, Celery, SQLAlchemy, Browser DevTools, Debugging, UI/UX Debugging, Automated Testing, Code Documentation, Structured Development Processes

Awards and Achievements

Completed Hacktoberfest 2024 with 4 open-source contributions, earning 4 HOLOPIN badges.

Soft Skills

- Strong analytical and problem-solving skills with a focus on debugging and troubleshooting.
- Quick learner with adaptability to new technologies and technical environments.
- Detail-oriented and process-driven with a focus on testing, documentation, and maintainable code.
- Collaborative team player with open-source contribution experience.